

수소활용소재개론

1주차: 수소의 전기화학 : 전기화학 개념
및 전극전위

국립부경대학교 금속공학전공

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1-1. 전기화학의 개념

Science

Physics : $A = B$

ex) $F = ma$
 $E = mgh$
.
.

Chemistry: $A \rightarrow B$ ex) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$

Electrochemistry



ex) $2\text{H}^+ + e^- \rightarrow \text{H}_2$
 $\text{Fe} \rightarrow \text{Fe}^{2+} + 2e^-$

1-1. 전기화학의 개념

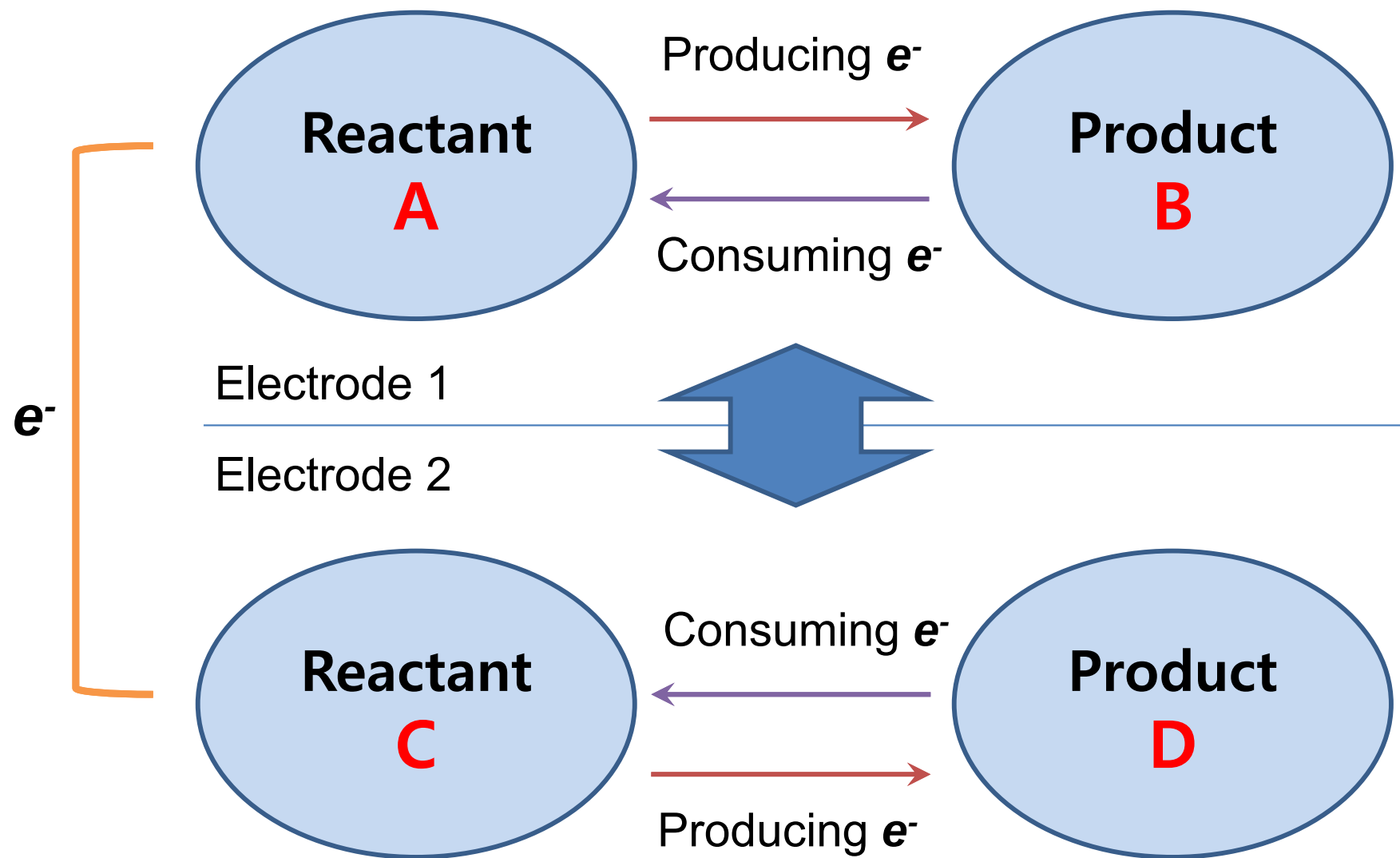
반응물 ?, 생성물 ?, 경로 ?

Chemical Reaction (화학반응)

Electrochemical Reaction (전기화학 반응)

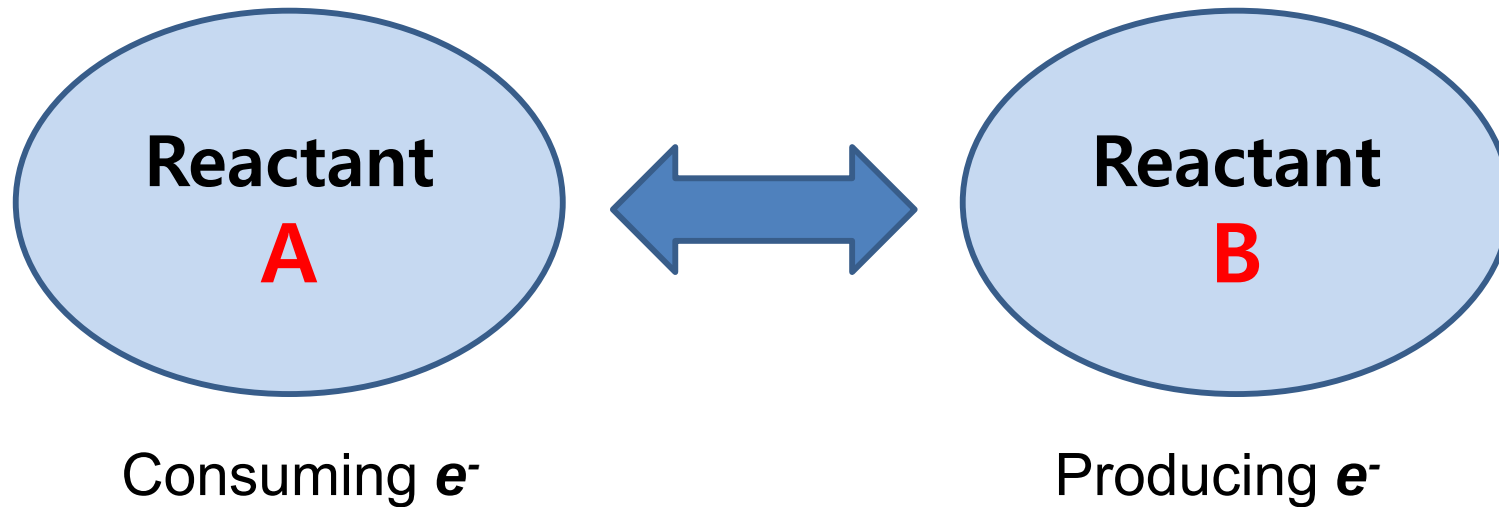
1-1. 전기화학의 개념

Electron



1-1. 전기화학의 개념

Ions



Producing anion (-)

Consuming cation (+)

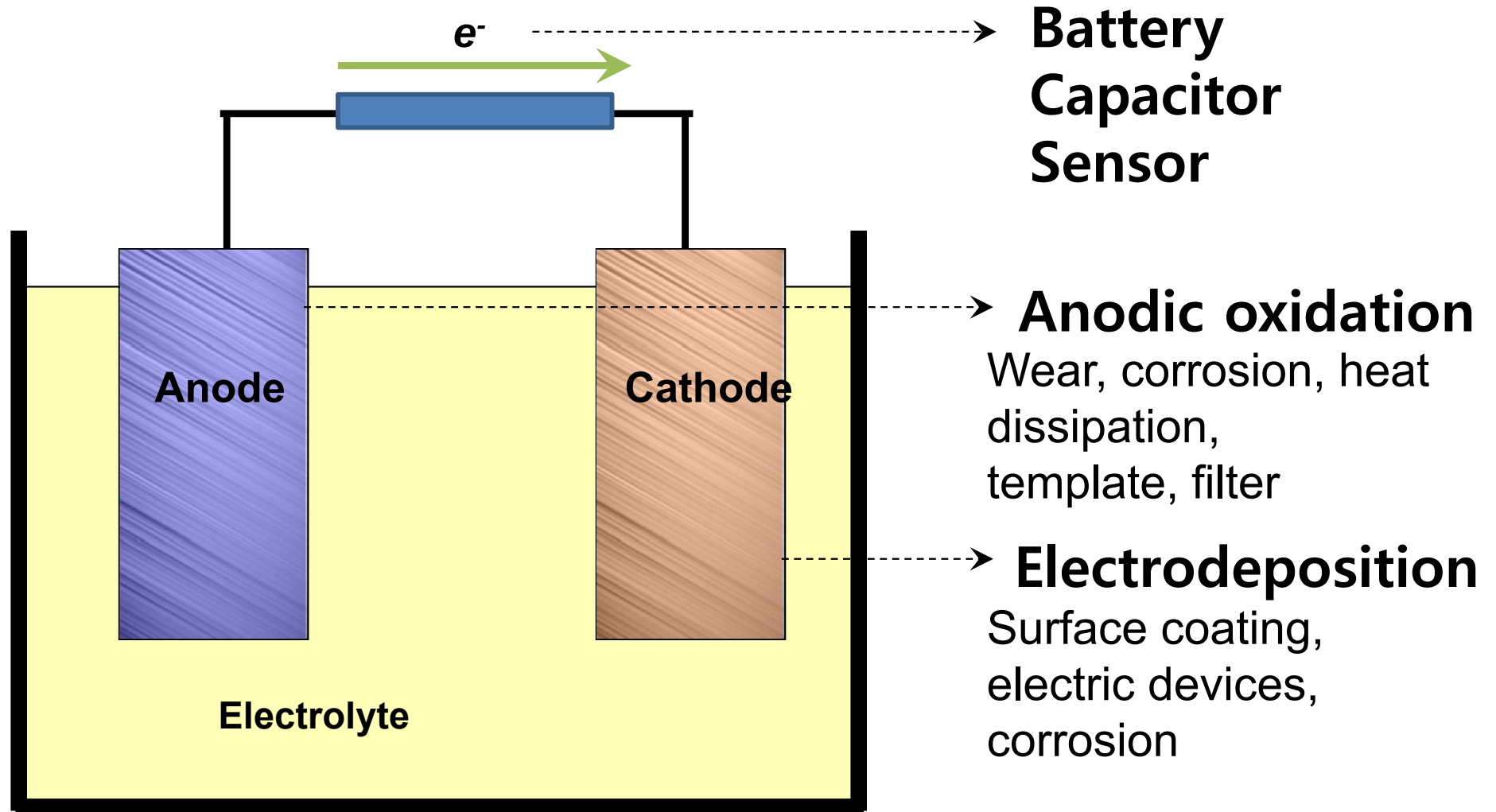
Producing cation (+)

Consuming anion (-)

Electrolyte

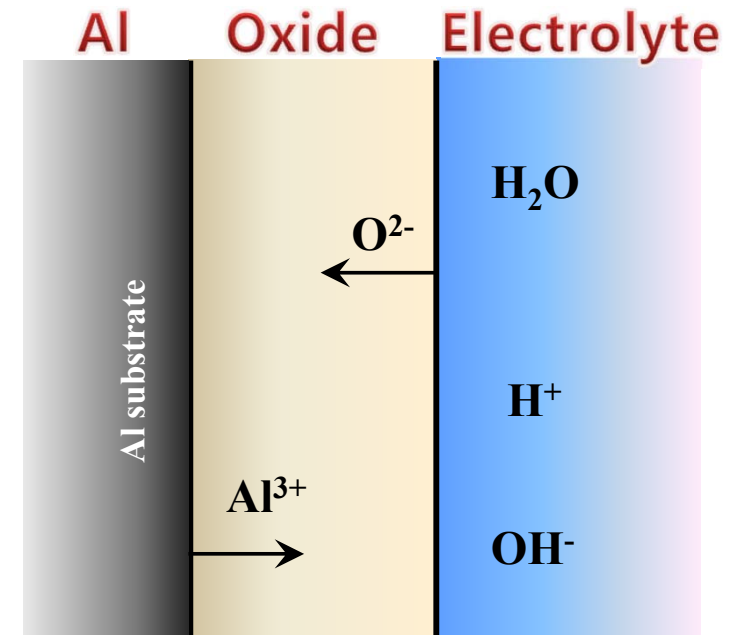
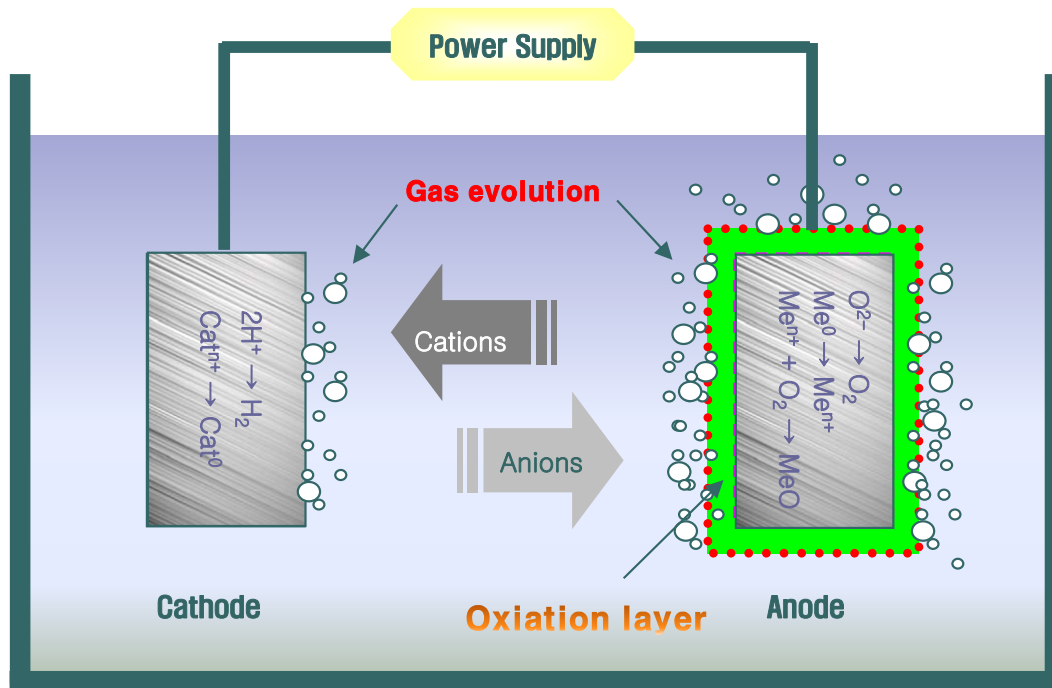
1-1. 전기화학의 개념

Ions



1-1. 전기화학의 개념

Anodic Oxidation

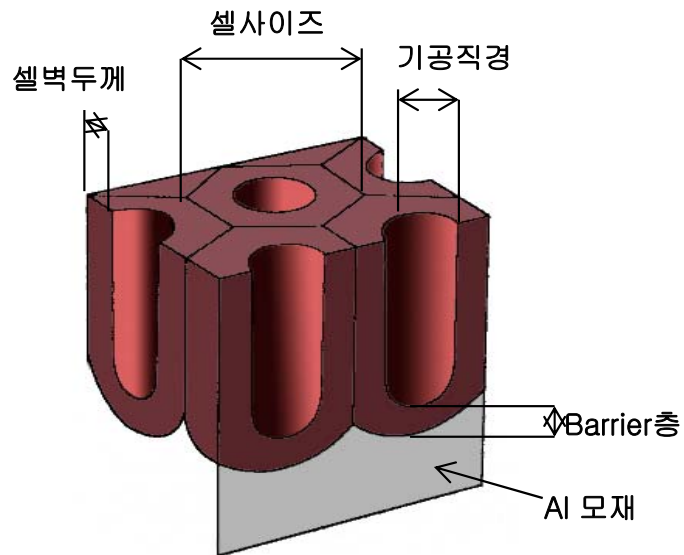


Overall reaction

- $2\text{Al} + 3\text{H}_2\text{O} \rightarrow \text{Al}_2\text{O}_3 + 3\text{H}_2$
- $2\text{Al} + 3\text{O}^{2-} \rightarrow \text{Al}_2\text{O}_3 + 6\text{e}^-$
- $2\text{Al}^{3+} + 3\text{H}_2\text{O} \rightarrow \text{Al}_2\text{O}_3 + 6\text{H}^+$
- $\text{OH}^- \rightarrow \text{H}^+ + \text{O}^{2-}$

1-1. 전기화학의 개념

Anodic Oxidation

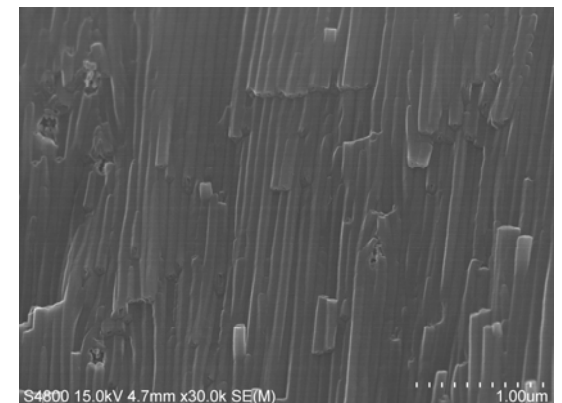
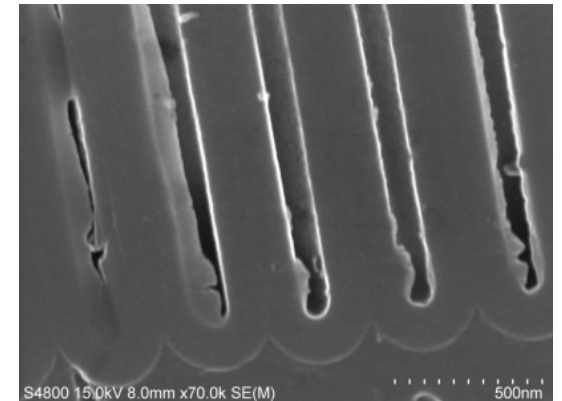
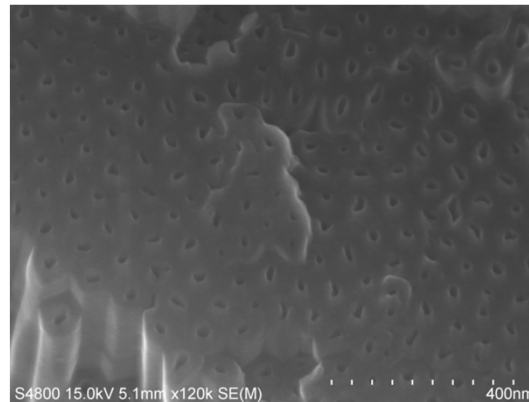
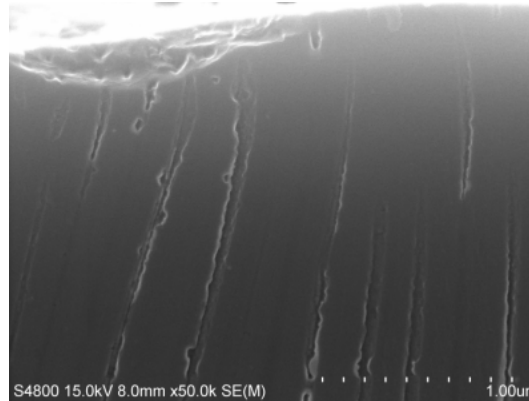


D_p : pore diameter

D_{int} : interpore distance

$t_{barrier}$: barrier layer thickness

$$\text{Porosity } P(\%) = (\pi/2\sqrt{3})(D_p/D_{int})^2$$



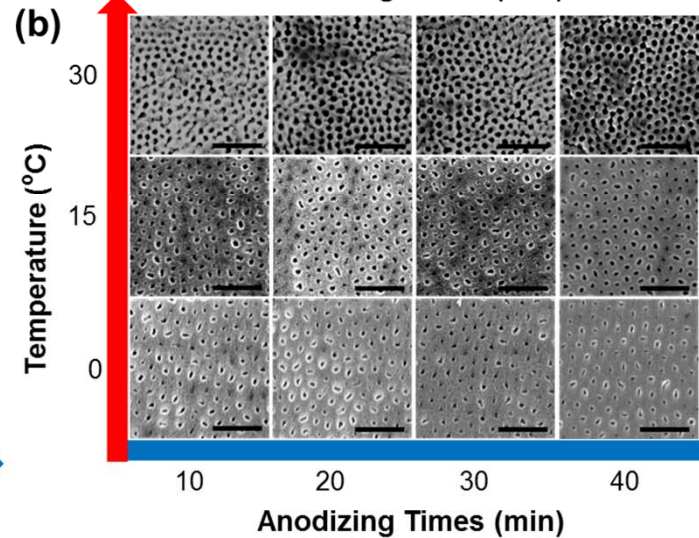
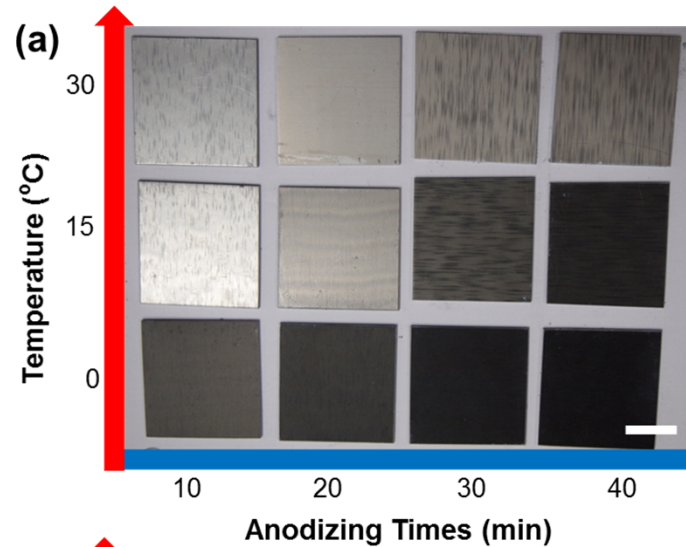
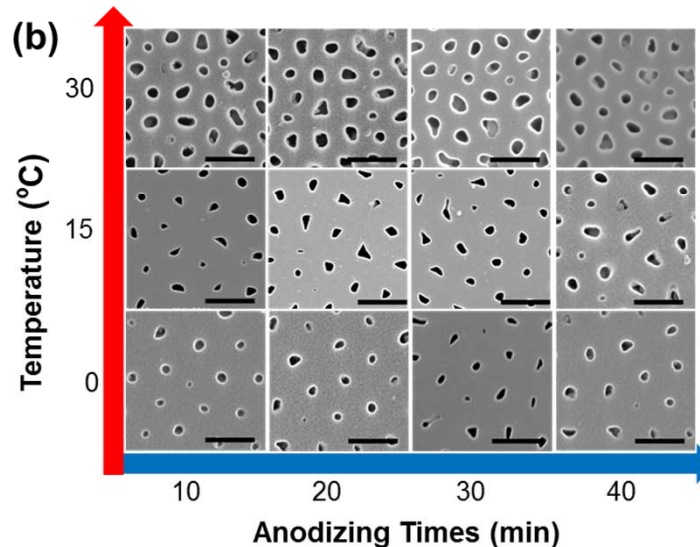
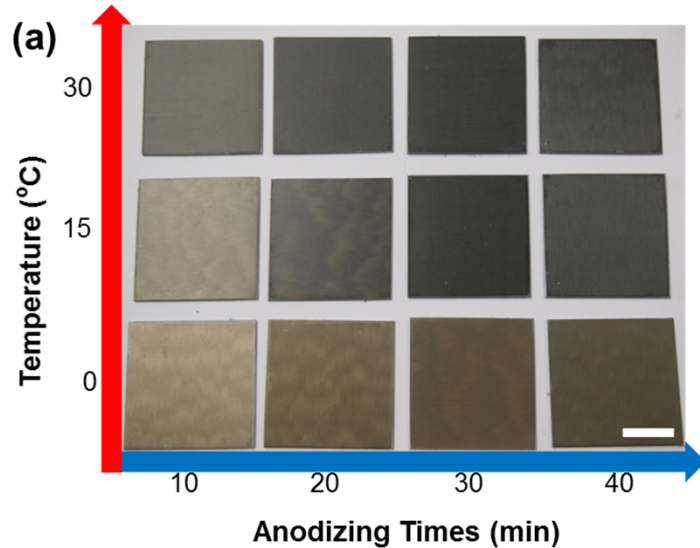
1-1. 전기화학의 개념

Anodic Oxidation

AAO from
Oxalic Acid

vs.

Sulfuric Acid



Pore shape and arrangement

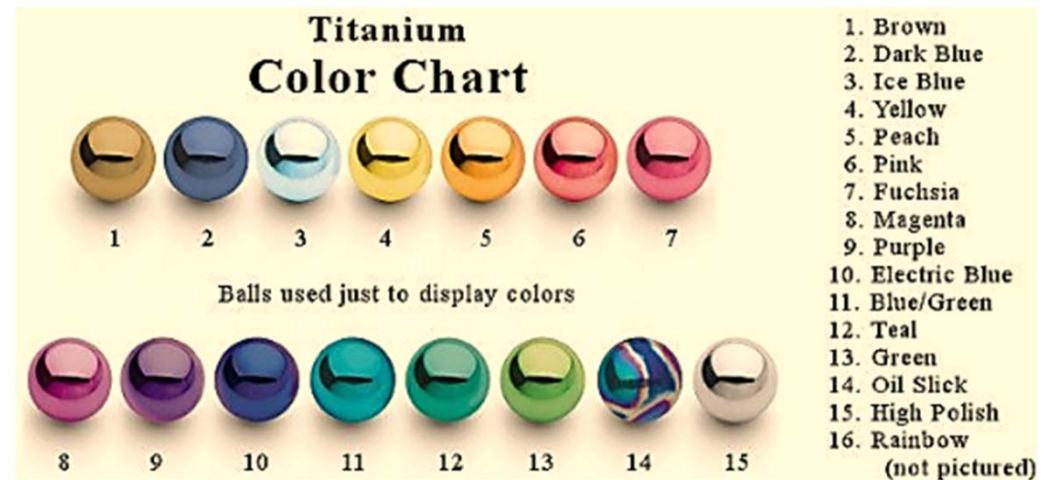
Oxalic acid:
irregular
pore arrangement,
pore shape and size.

Sulfuric acid:
Relatively ordered
pore arrangement,
regular
Pore shape and size

No effect of thickness
on pore arrangement
size and shape

1-1. 전기화학의 개념

Anodic Oxidation



1-1. 전기화학의 개념

Corrosion



(www.roplant.org)



1-1. 전기화학의 개념

Electrodeposition of Zinc

차량의 부식



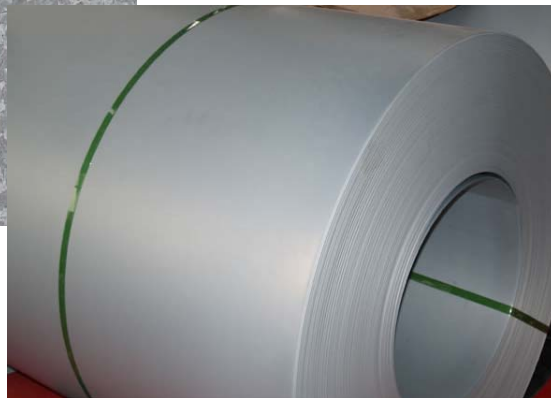
Rust-Belt



아연도금을 이용한 방식



Zinc deposit

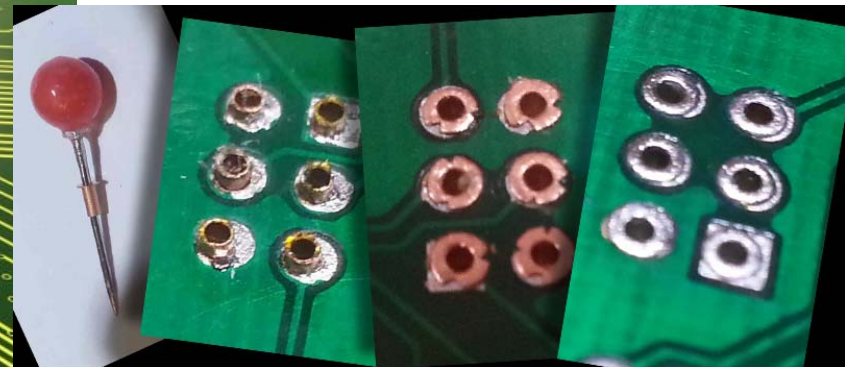
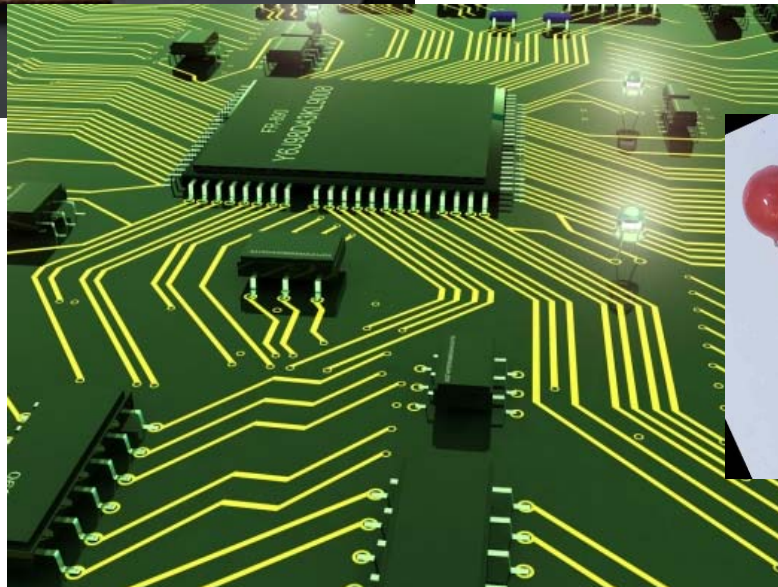
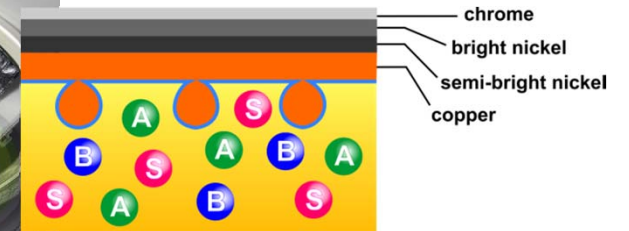
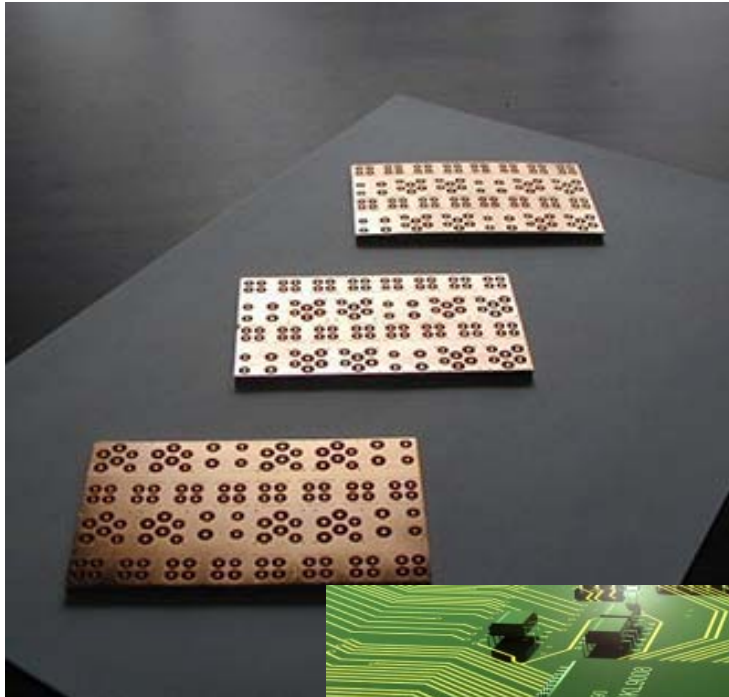


Steel coil



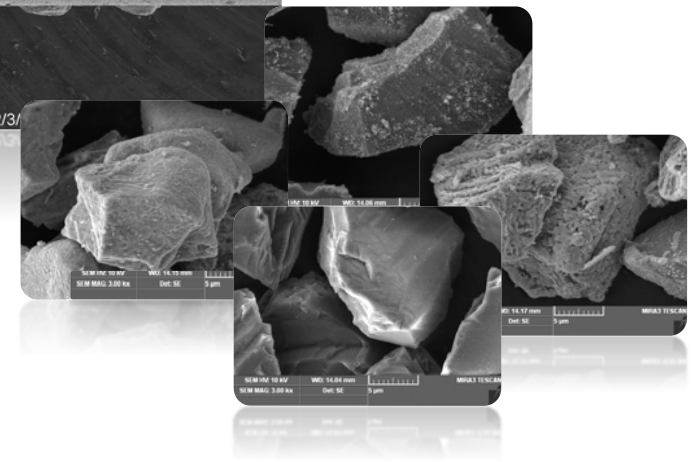
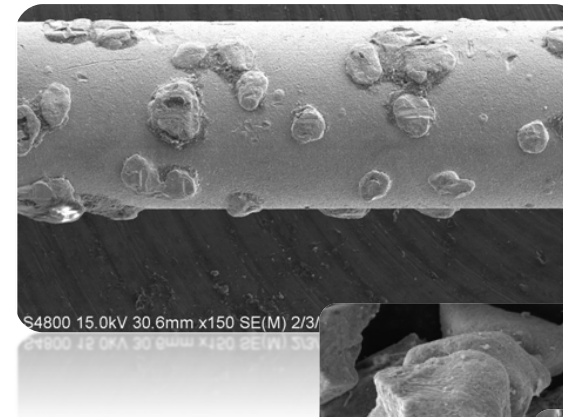
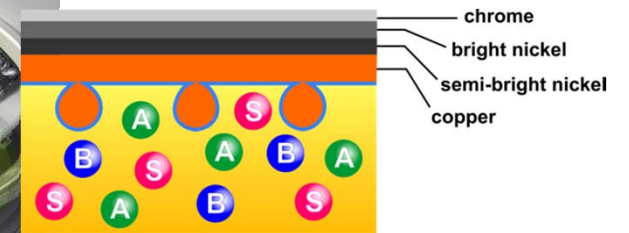
1-1. 전기화학의 개념

Electrodeposition of Copper



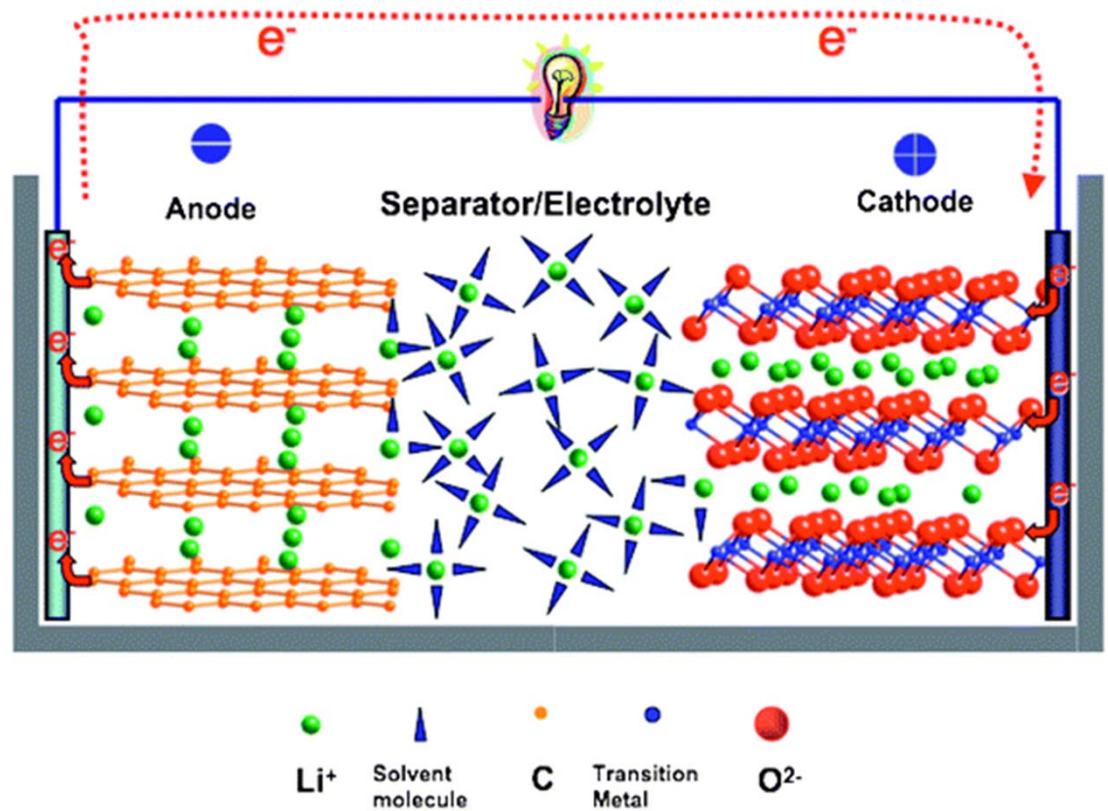
1-1. 전기화학의 개념

Electrodeposition of Nickel



1-1. 전기화학의 개념

Battery

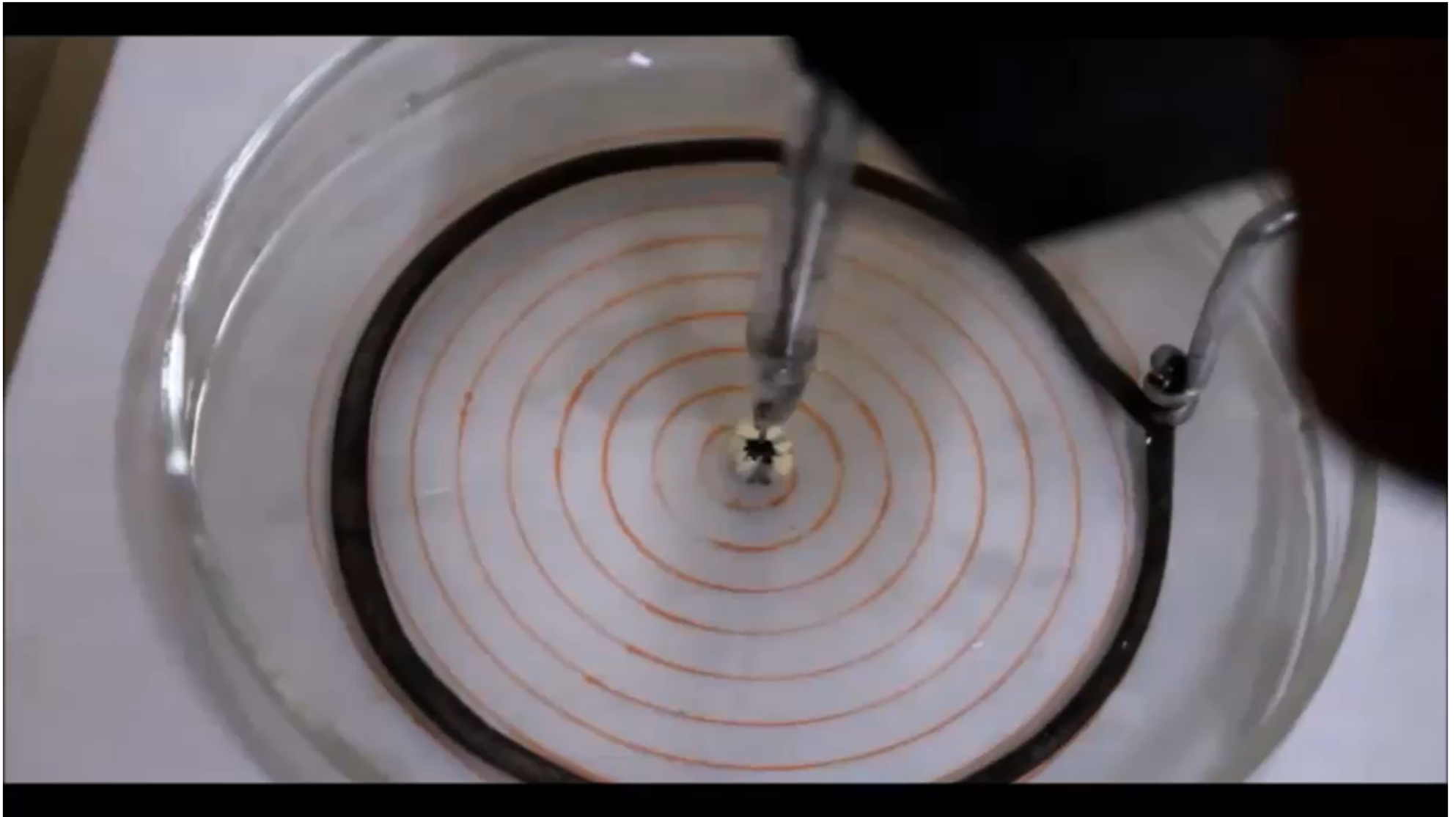


1-1. 전기화학의 개념

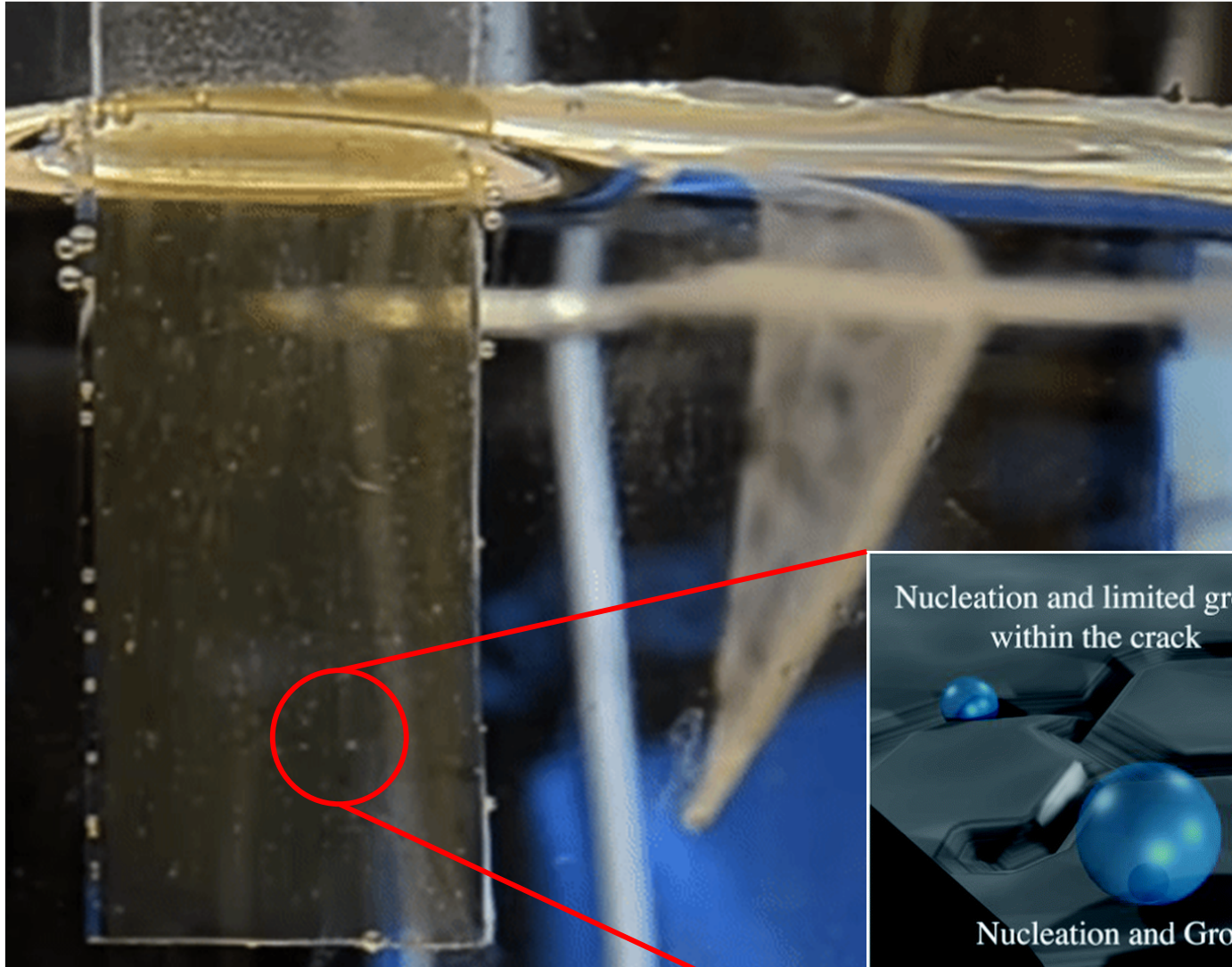
Fuel Cell



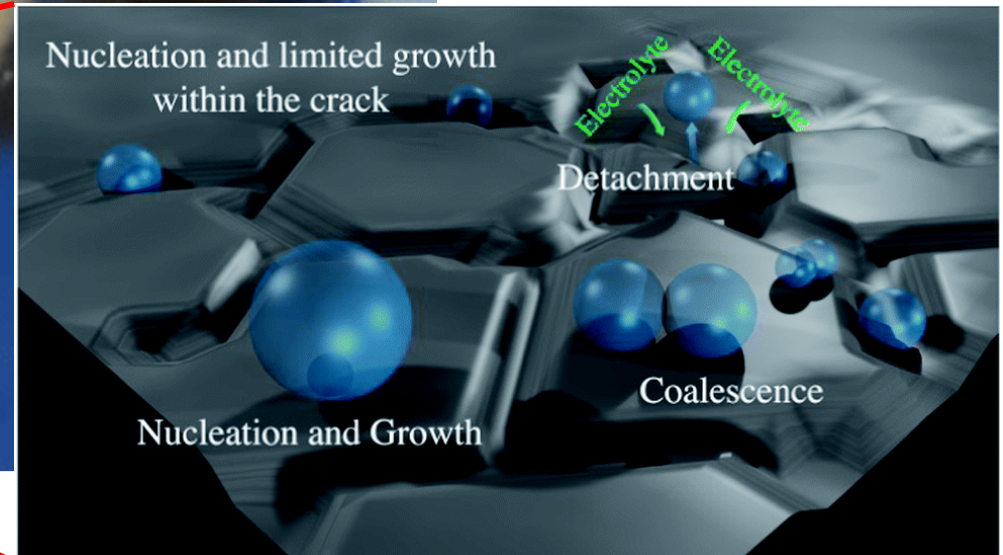
1-2. 전극 전위



1-2. 전극 전위

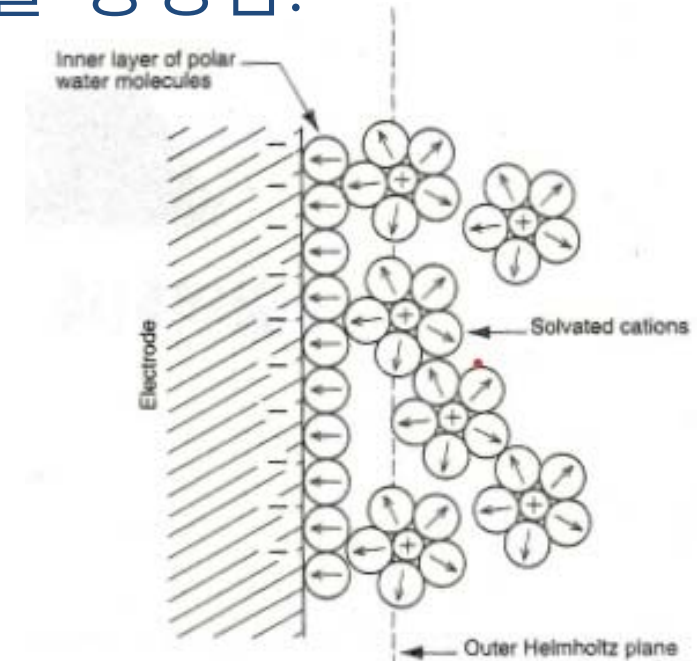
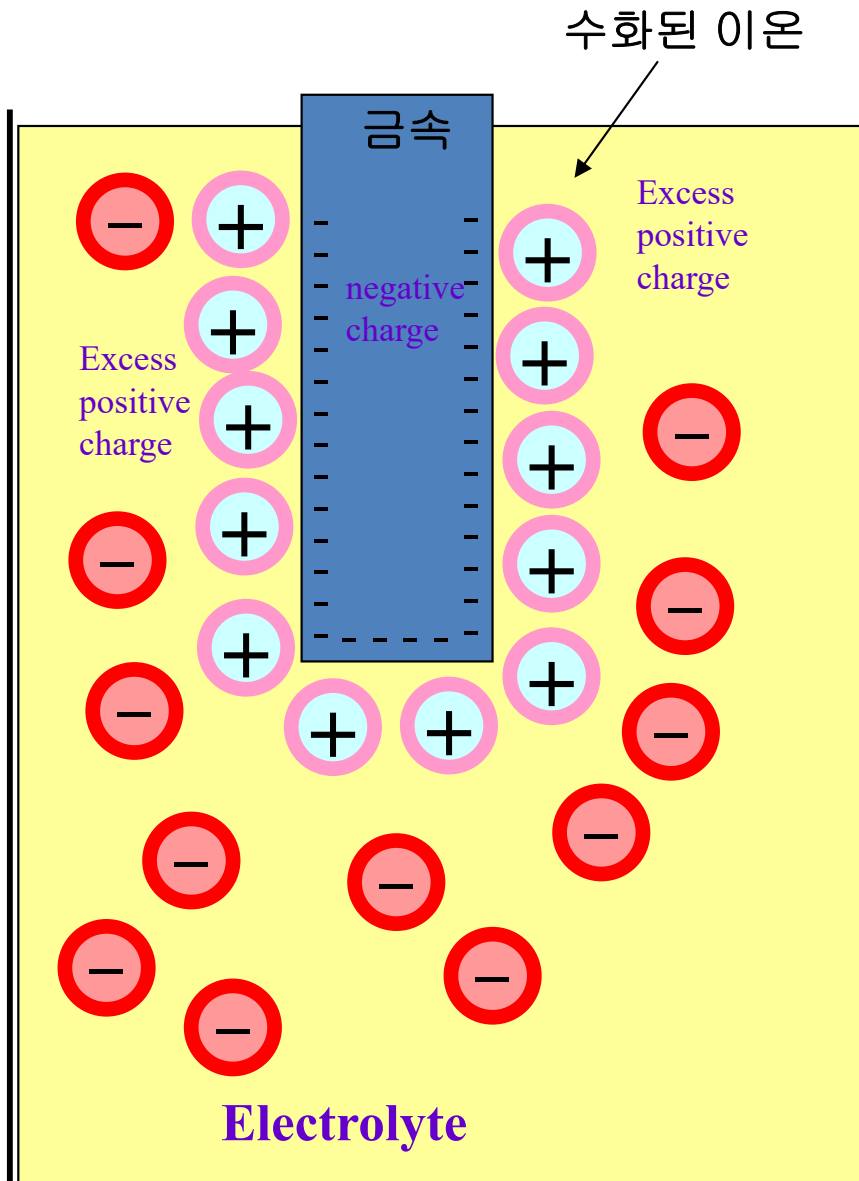


What happen?



1-2. 전극 전위

금속 전극은 전해질에서 전기적 이중층을 형성함.



금속 전극의 산화로 양이온을 생성함. 반응 결과로 잉여 전자들이 금속 전극에 누적됨.

금속 표면에 자유전자와 전해질내의 양이온들 사이에 인력이 발생함.

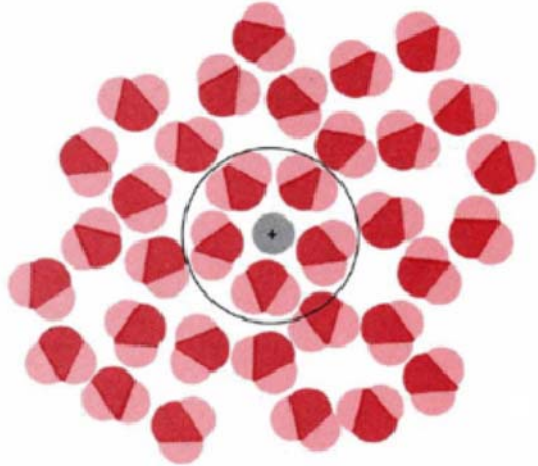
전극과 전해질 사이에 음전하층과 양전하층의 형성
전기적 이중층 (Electrical double layer)가 형성.

계면 전위차에 의해 반응이 발생함

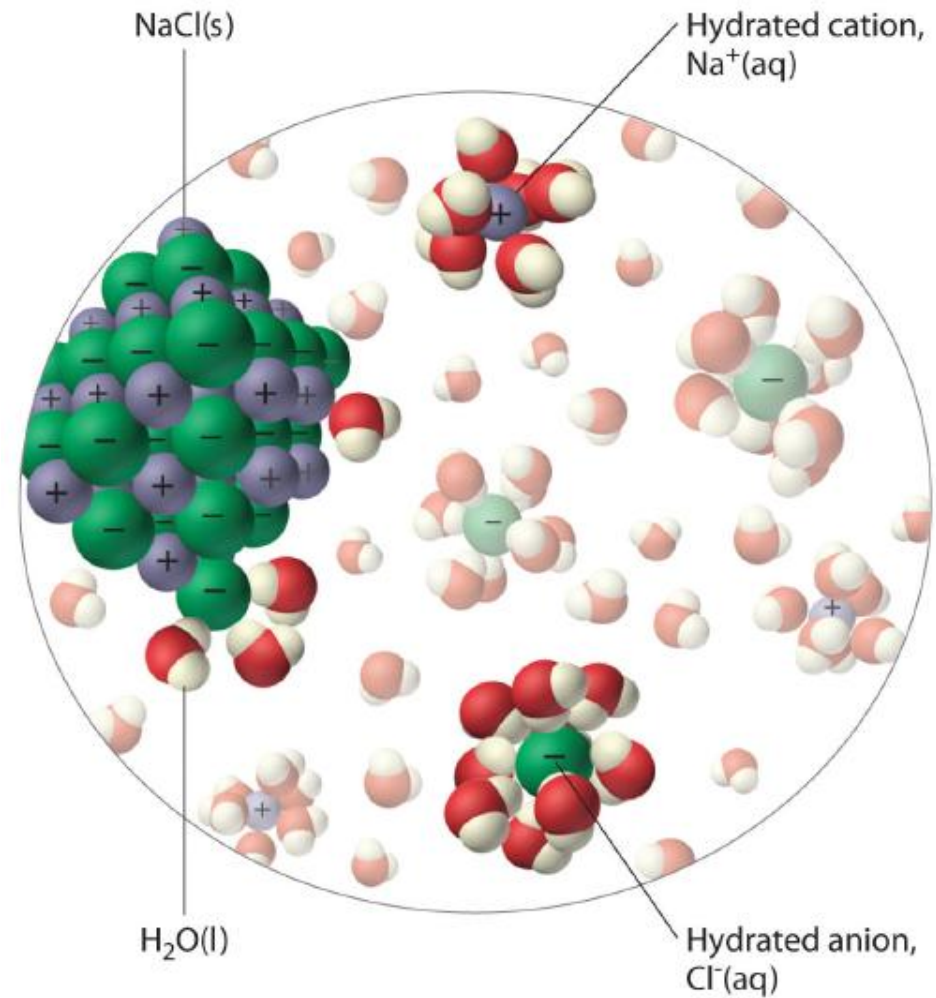
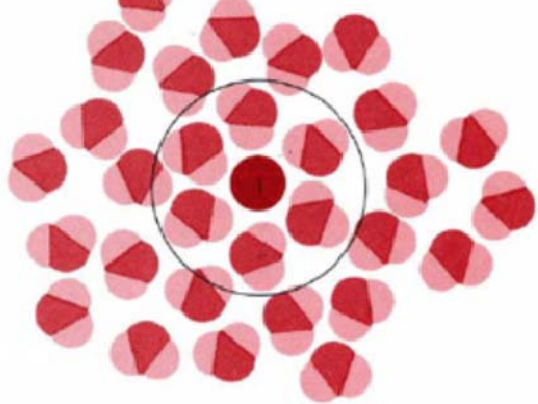
1-2. 전극 전위

물 속에서 이온은 수화됨.

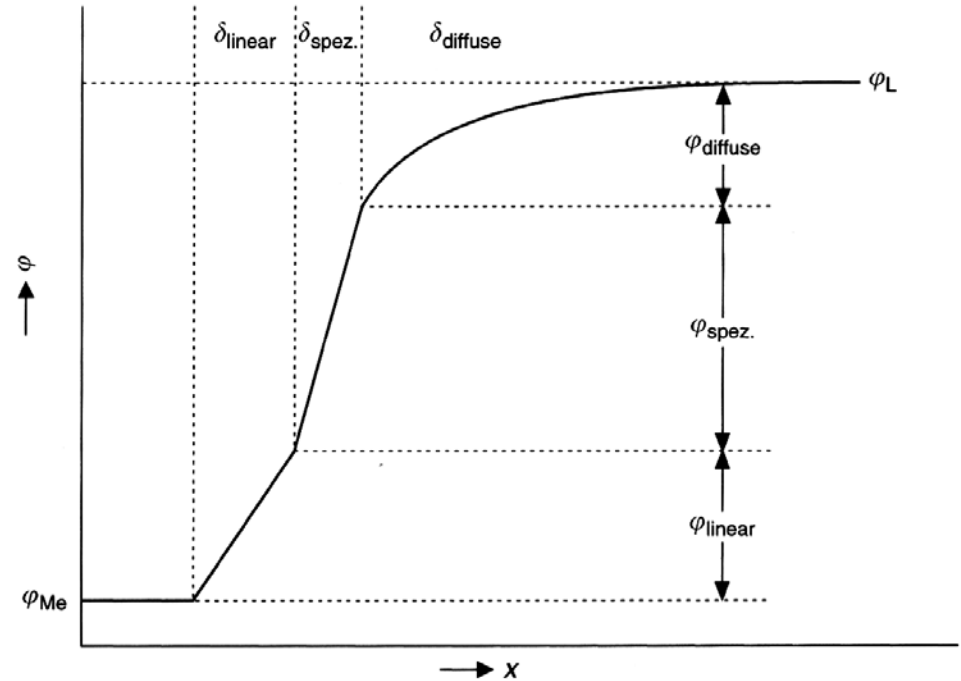
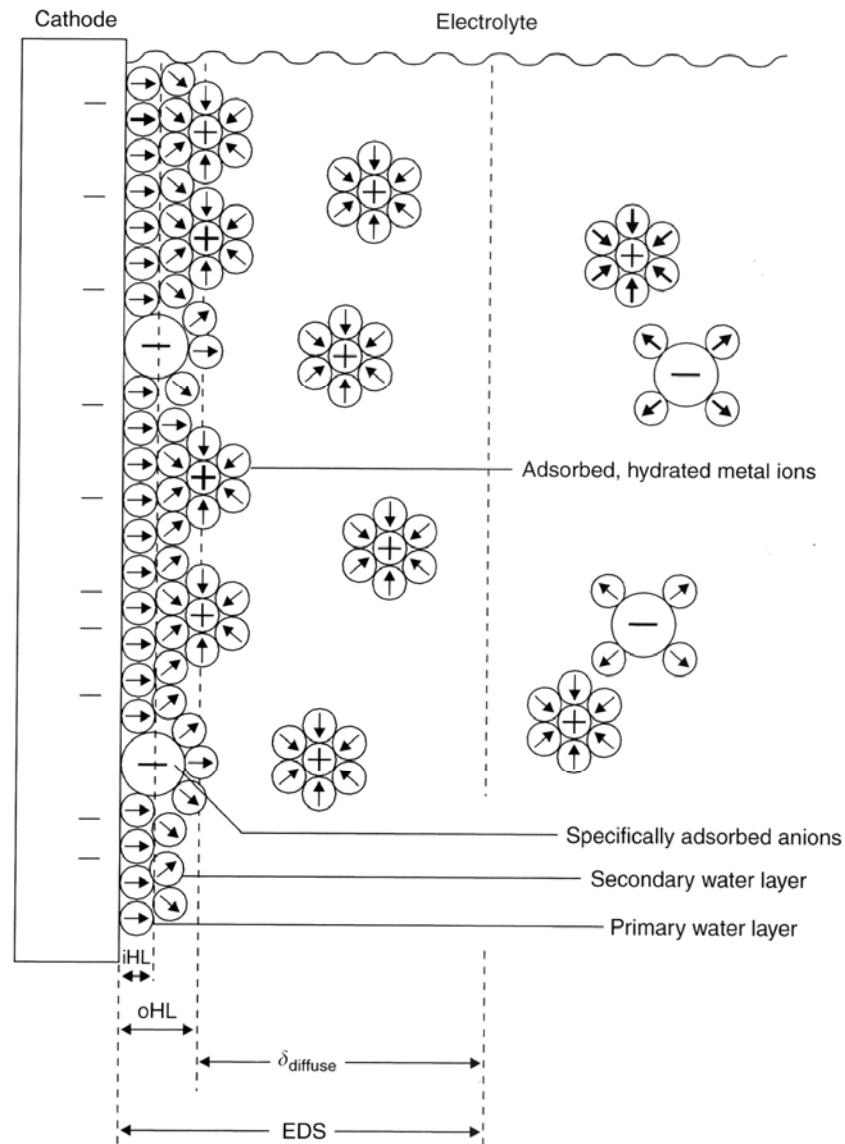
양이온(Cation)



음이온(Anion)

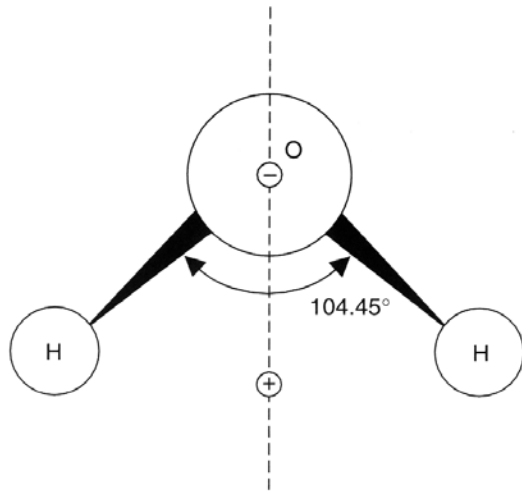


1-2. 전극 전위

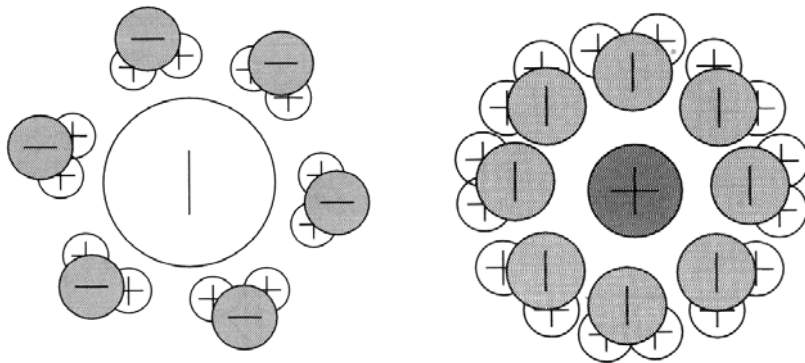


[Stern-Graham model]

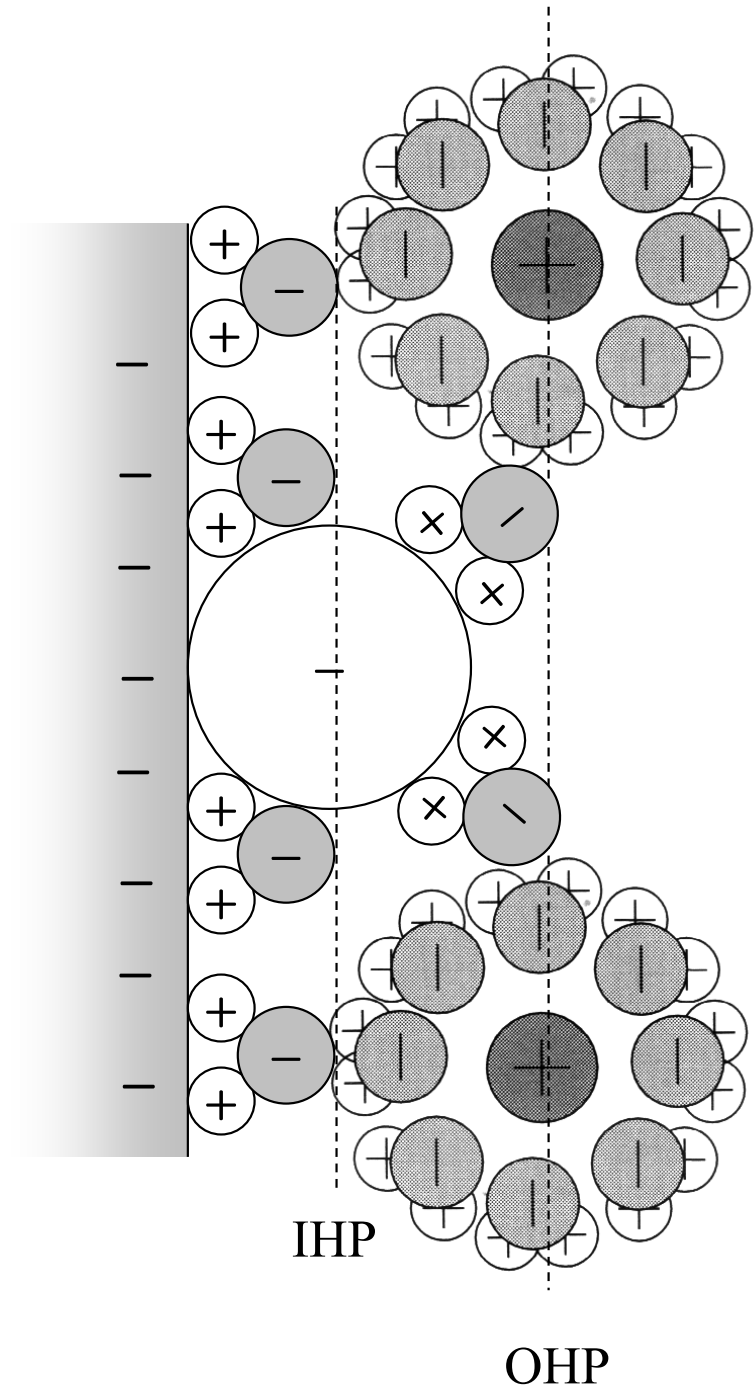
1-2. 전극 전위



The structure of a water molecule-dipole

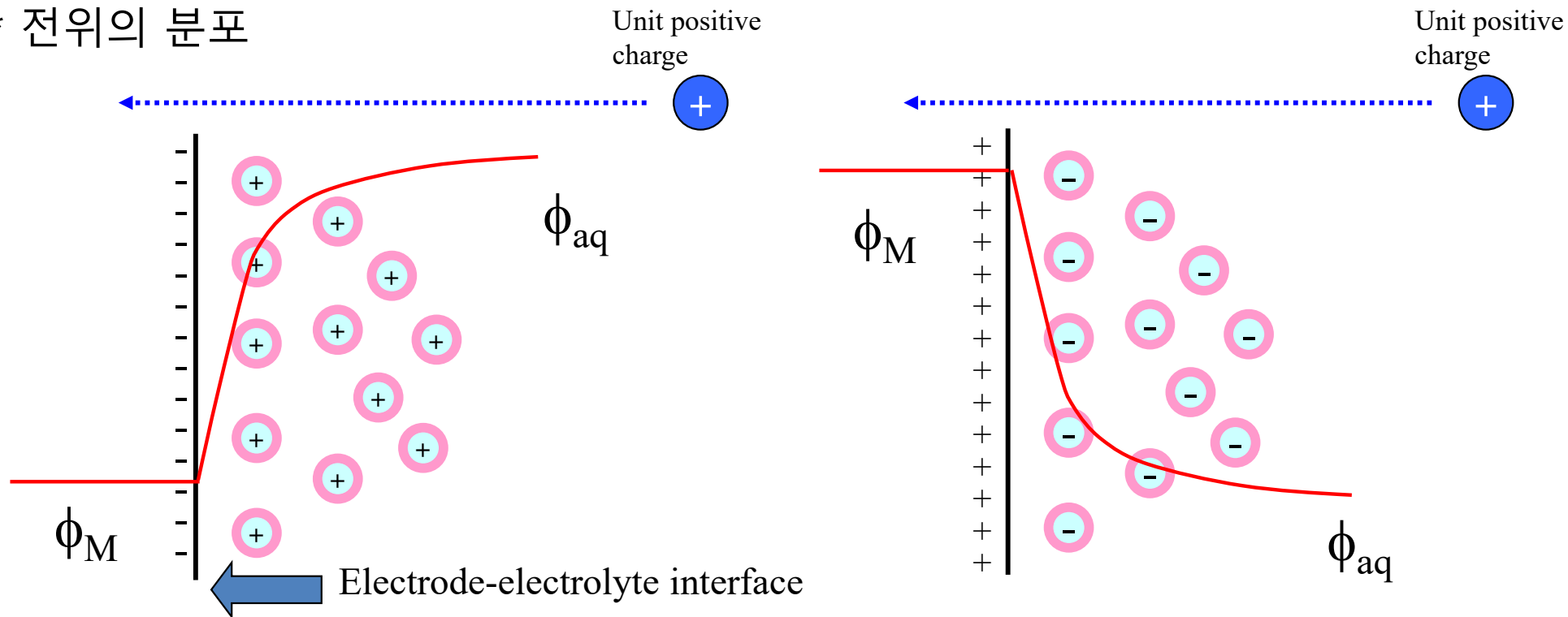


Hydrated cation and anion

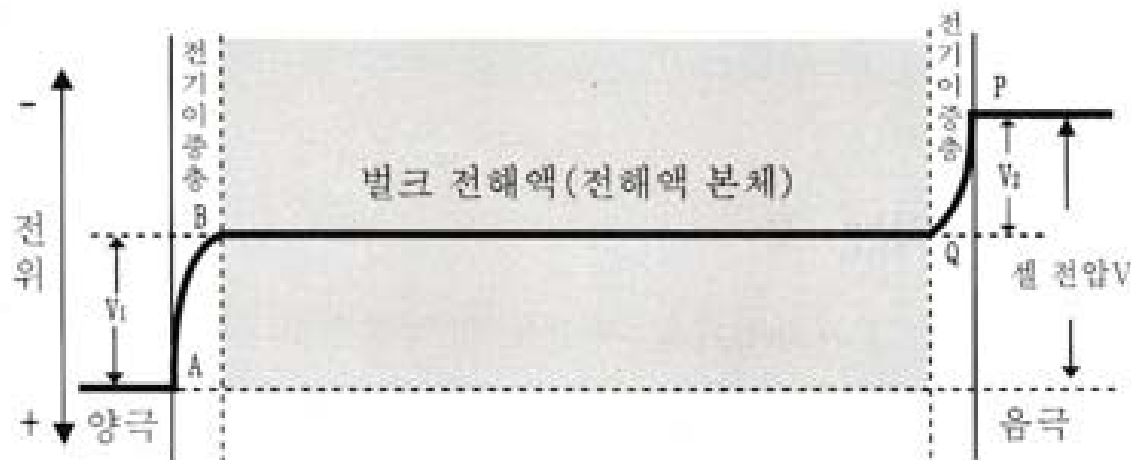


1-2. 전극 전위

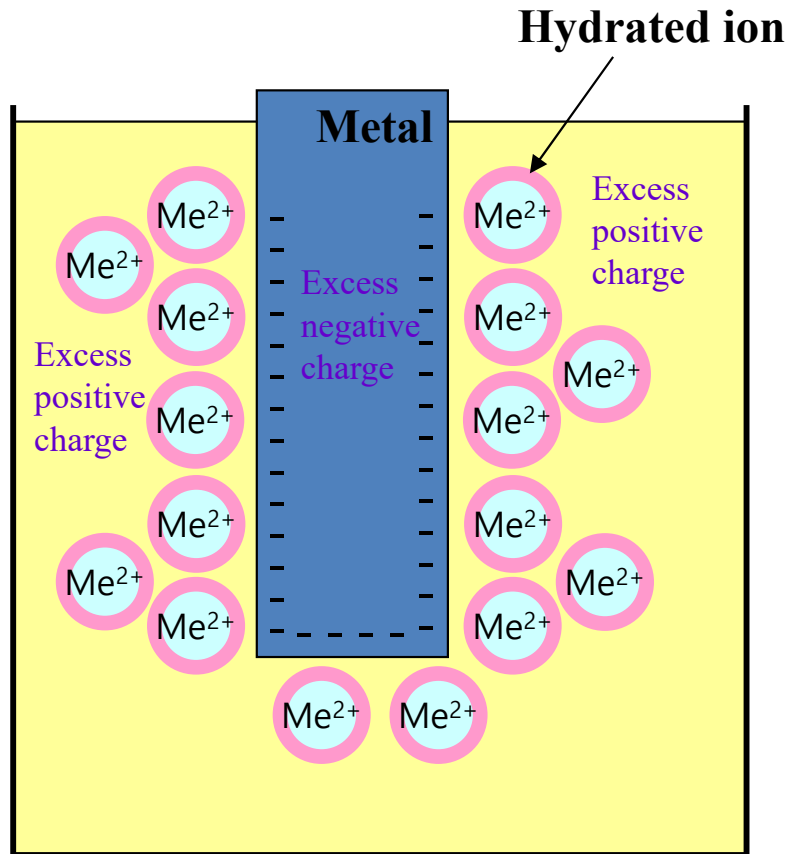
* 전위의 분포



Interfacial potential difference → absolute electrode potential ($=\phi_M - \phi_{aq}$)



1-3. 전극 전위와 ΔG

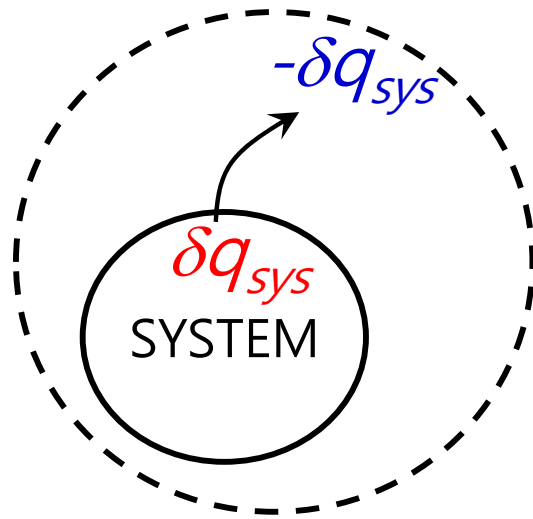


Reduction rate = Oxidation rate $\gg$ equilibrium

Gibb's Free Energy (ΔG)

열역학적 평형상태로부터
전극전위에 대한 정보를 구할
수 있음.

1-3. 전극 전위와 ΔG



자발적 반응이
되기 위한 조건

$$\begin{aligned} dS_{tot} &= dS_{sys} + dS_{sur} = dS_{sys} + \frac{\delta q_{sur}}{T} \\ &= dS_{sys} - \frac{\delta q_{sys}}{T} > 0 \end{aligned}$$

자발적 반응이 일어나기 위한 조건은

$$\Delta G_{T,P} = \Delta H_{sys} - T\Delta S_{sys} < 0$$

→ 시스템의 온도와 압력이 일정할 때,

시스템의 내부 엔탈피 변화와 시스템의 엔트로피 변화를 관찰하면,

주변의 엔트로피 변화를 관찰하지 않고도, 반응의 자발성을 판정할 수 있다.

→ Gibbs Free Energy

1-3. 전극 전위와 ΔG

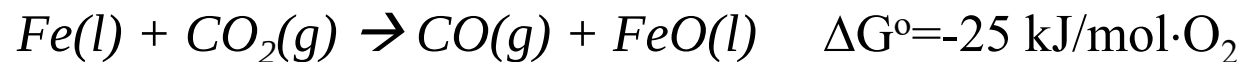
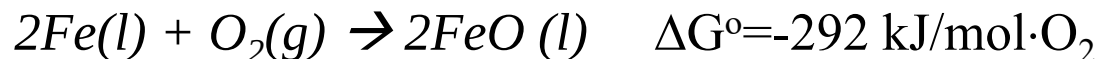
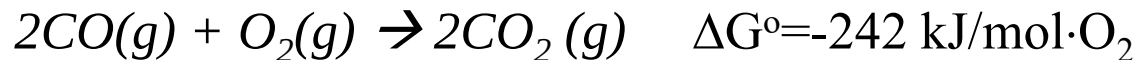
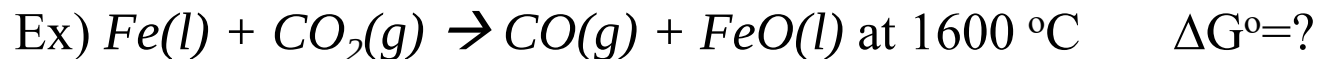
ΔG 를 결정하는 방법

(1) $\Delta G_T^o = \Delta H_T^o - T\Delta S_T^o$ 에서 enthalpy 와 entropy 를 독립적으로 결정

(2) 각 성분의 “표준 생성 자유에너지” 변화를 바탕으로 아래의 식을 이용하여 결정

$$\Delta G_T^o = \sum_{\text{Products}} \nu G_T^o - \sum_{\text{Reactants}} \nu G_T^o$$

(3) 기존의 “표준 반응 자유에너지” 정보를 바탕으로, Hess's law 를 이용하여 결정



1-3. 전극 전위와 ΔG

ΔG 를 결정하는 방법

(4) 표준 상태에서 벗어나는 경우, 압력 항을 보정하면, 일반적인 상황에서의 ΔG -T plot 을 얻을 수 있다.

$$\int_{G_A^o}^{G_A} dG = \int_{\substack{p_A^o \\ \text{1 bar}}}^{p_A} \frac{RT}{p} dp \quad \Rightarrow \quad G_A = G_A^o + RT \ln p_A$$

$$\text{For } aA + bB = cC + dD \quad \Delta G = \Delta G^o + RT \ln \frac{p_C^c p_D^d}{p_A^a p_B^b}$$

Example)

$$2M(s) + O_2(g) = 2MO(s) \quad \Delta G = \Delta G^o + RT \ln \frac{\overset{\text{1 (for pure substance)}}{a_{MO}^2}}{a_M^2 p_{O_2}} \quad \Rightarrow \quad \Delta G = \Delta G^o - RT \ln p_{O_2}$$

따라서, 평형 상태 ($\Delta G=0$) 에서 $\Delta G^o = RT \ln p_{O_2}$

1-3. 전극 전위와 ΔG

Interfacial potential difference $\rightarrow$ **absolute electrode potential ($=\phi_M - \phi_{aq}$)**

Concentration of electrons
in Electrode

Concentration of cation
in Electrolyte



Electrolyte Electrode

Reduction rate = Oxidation rate $\gg$ equilibrium

Gibb's Free Energy (ΔG)

1-3. 전극 전위와 ΔG

Electrochemical Potential



Electrode Potential

$$\Delta G^{\circ} = -nFE^{\circ} \quad (\text{standard conditions})$$

↓

$$\Delta G = -nFE \quad (\text{non-standard conditions})$$

n : valence electron

F : Faraday Constant (96500C/mol)

☞ If the reaction involves charge (or electron) transfer, the chemical potential (ΔG) can be transformed to the electrochemical potential (E).

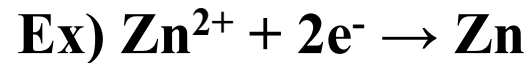
1-3. 전극 전위와 ΔG

$$\Delta G^{\circ} = -nFE^{\circ} \text{ (표준상태)}$$

$$\Delta G = -nFE \text{ (비 표준상태)}$$

n : 반응에 참여하는 전자의 수

F : Faraday's Constant (96500C/mol)



$$\Delta G^{\circ} = 146688 \text{ J/mol}$$

$$\begin{aligned} E^{\circ} &= \frac{-146688}{2 \times 96500} \\ &= \underline{-0.76 \text{ V}} \end{aligned}$$

Nernst Equation 을 통하여 전자를 포함하는 반응의 화학적 포텐셜을 전기화학적 포텐셜로 바꿀 수 있다.

1-3. 전극 전위와 ΔG

원소	환원 반쪽 반응		표준 환원 전위 E^0 , volts
Li	$\text{Li}^+ + e^-$	$\longrightarrow \text{Li}$	-3.045
K	$\text{K}^+ + e^-$	$\longrightarrow \text{K}$	-2.925
Ca	$\text{Ca}^{2+} + 2e^-$	$\longrightarrow \text{Ca}$	-2.87
Na	$\text{Na}^+ + e^-$	$\longrightarrow \text{Na}$	-2.714
Mg	$\text{Mg}^{2+} + 2e^-$	$\longrightarrow \text{Mg}$	-2.37
Al	$\text{Al}^{3+} + 3e^-$	$\longrightarrow \text{Al}$	-1.66
Zn	$\text{Zn}^{2+} + 2e^-$	$\longrightarrow \text{Zn}$	-0.763
Cr	$\text{Cr}^{3+} + 3e^-$	$\longrightarrow \text{Cr}$	-0.74
Fe	$\text{Fe}^{2+} + 2e^-$	$\longrightarrow \text{Fe}$	-0.44
Cd	$\text{Cd}^{2+} + 2e^-$	$\longrightarrow \text{Cd}$	-0.403
Ni	$\text{Ni}^{2+} + 2e^-$	$\longrightarrow \text{Ni}$	-0.25
Sn	$\text{Sn}^{2+} + 2e^-$	$\longrightarrow \text{Sn}$	-0.14
Pb	$\text{Pb}^{2+} + 2e^-$	$\longrightarrow \text{Pb}$	-0.126
H ₂	$2\text{H}^+ + 2e^-$	$\longrightarrow \text{H}_2$	0.000 (기준 전극)
Cu	$\text{Cu}^{2+} + 2e^-$	$\longrightarrow \text{Cu}$	+0.337
I ₂	$\text{I}_2 + 2e^-$	$\longrightarrow 2\text{I}^-$	+0.535
Hg	$\text{Hg}^{2+} + 2e^-$	$\longrightarrow \text{Hg}$	+0.789
Ag	$\text{Ag}^+ + e^-$	$\longrightarrow \text{Ag}$	+0.799
Br ₂	$\text{Br}_2 + 2e^-$	$\longrightarrow 2\text{Br}^-$	+1.08
Cl ₂	$\text{Cl}_2 + 2e^-$	$\longrightarrow 2\text{Cl}^-$	+1.360
Au	$\text{Au}^{3+} + 3e^-$	$\longrightarrow \text{Au}$	+1.50
F ₂	$\text{F}_2 + 2e^-$	$\longrightarrow 2\text{F}^-$	+2.87

산화제의 세기: 환원의 용이함

환원제의 세기: 산화의 용이함